

Computational Screening of Redox-Active Monomers

A physics-based, multi-objective candidate-selection pipeline

RTFB Materials Research

October 2, 2026

Redox potential · Stability · Kinetics · Capacity

The goal: pick good redox molecules for flow batteries

System. A **Merrifield-resin polymer** carrying pendant redox groups. The redox chemistry is *local*, so we model the **monomer as a proxy**: establish we can capture the local chemistry, then add backbone effects.

Selection is multi-objective. Voltage and stability are necessary but not sufficient:

$$\text{Energy density} \propto n \cdot V_{\text{cell}} \cdot \text{solubility}$$

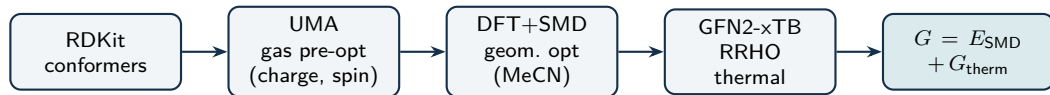
- **redox potential** $\rightarrow$ cell voltage / window
- **stability** $\rightarrow$ cycle life
- **kinetics** (λ), **capacity** (n), cost, crossover

Guiding principle

Energy *differences* with error cancellation compute **reliably**; *absolute* charged-species solvation does **not**.

Build objectives on the cancellation-friendly quantities; treat the rest as labelled proxies or filters.

The pipeline (per redox state)



Level of theory

- opt: r²SCAN/def2-SVP(D) + SMD
- energy: ω B97M-V/def2-TZVPD (all charges)
- referenced to level-matched Fc/Fc⁺

Two hard-won fixes

- gpu4pyscf open-shell Hessian is **broken** → thermal from **GFN2-xTB** RRHO
- cross-points gas-only (do_smd=False), ~2× cheaper

How each axis is computed

axis	method	validation / trust
redox potential	DFT+SMD ΔG ; ω B97M-V/def2-TZVPD // r ² SCAN/def2-SVP(D); level-matched Fc live	OROP: MAE 0.52 V; within-class $\rho = 0.78/0.79$
thermal G_{therm} stability (disprop.)	GFN2-xTB RRHO (298 K) $\Delta G_{\text{disp}} = F(E_{\text{hi}} - E_{\text{lo}})$ from state G 's	open-shell safe 10 sourced spacings: ρ 0.80, over-est. +23 kJ/mol
stability (dimer.) reorg. energy λ	pimer cluster DFT+SMD ΔG $\lambda_O + \lambda_R$ (Nelsen); molecular-cavity PCM λ_o	flagged (+2 desolvation) D3TaLES: median $ \Delta = 0.09$ eV ($n = 322$)
integrity gate	bond-graph change + solvated EA	necessary, not sufficient for reversibility
capacity	$n \cdot F/\text{MW}$ (accessible n)	exact
solubility	$\Delta G_{\text{solv}} = E_{\text{SMD}} - E_{\text{gas}}$ (neutral)	proxy (relative)
synth. access.	RDKit SA_Score	heuristic proxy

Geometries optimized in SMD(MeCN); conformers via RDKit + UMA pre-opt.

Redox potential: benchmarked vs. experiment (OROP)

Production-protocol OROP benchmark ($n = 169$):

	ours	OROP impl.-DFT
MAE (V)	0.52	0.28
signed bias	+0.49	—
Spearman (all)	0.92	—

At OROP's own level our organic potentials match theirs (~ 0.07 V); the gap sits in the Fc reference (4.19 vs 4.66 V). Sourced MeCN set: E1 bias +0.26 V, E2 -0.07 V. **Not calibrated away.**

Ranking must be judged *within* charge class

class	n	MAE	Spearman
+1/0 couples	118	0.48	0.79
0/−1 couples	49	0.60	0.78

Global correlation is inflated because charge classes separate. Ranking is assessed within each class; the integrity gate removes category errors.

The anion “failure” was category errors, not bad physics

Some apparent anion failures are category errors: molecules with **no intact, bound reduced state**:

molecule	exp	RMSD _{neu→an}
CCl ₄	-2.57	3.4 Å
CH ₂ Br ₂	-3.12	1.3 Å
(unbound anions)	—	EA _{gas} < 0

Dissociative electron attachment
(CCl₄+e⁻→CCl₃[•]+Cl⁻) — scored against irreversible *peak* potentials.

Integrity screen

Bound anion (solvated EA > 0) *and* intact (**bond-graph unchanged**, not raw RMSD — which is confounded by tether flexibility).

- filtering → anion Spearman **0.49**→**0.95**
- a required necessary-condition gate and stability signal
- does **not** test proton transfer, dimerization, or long-term cycling

Stability = decomposition free energies

Disproportionation $2R \rightarrow R^+ + R^-$

(atom- & electron-conserving; ranks well, ρ 0.80, but **over-estimated by ~ 23 kJ/mol**):

intermediate	ΔG_{disp} (kJ/mol)
TEMPO $\bullet$	+228
AQ semiquinone	+82
viologen MV $^{+\bullet}$	+55

TEMPO — the textbook persistent radical (our validation anchor) — ranks most stable, as it must. Exact from existing energies; no new calculation.

Dimerization $2 \text{MV}^{+\bullet} \rightarrow [\text{MV}_2]^{2+}$ (the real viologen failure mode):

$$\Delta G_{\text{dim}} = +0.49 \text{ eV (flagged)}$$

Config sampling: minimum well-defined ($\sigma_{\text{config}} \approx 2 \text{ meV}$).

The uncertainty is the **+2 pimer desolvation** — the continuum's weak spot.

$\Rightarrow$ use dimerization as a **relative** ranking; lean on disproportionation for **ranking** (absolute values over-estimated, FINDINGS #24).

Kinetics: inner- and outer-sphere reorganization

Inner-sphere λ_i — solute geometry relaxation,
4-point (Nelsen):

$$\lambda_i = [E_O(q_R) - E_O(q_O)] + [E_R(q_O) - E_R(q_R)]$$

gas single points; same four-point formula as D3TaLES. **Validated.**

Outer-sphere λ_o — solvent reorganization
(Marcus/Born, **Sharma Eq. 13**):

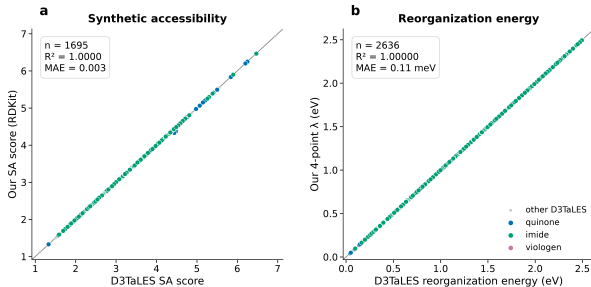
$$\lambda_o = \frac{z^2 e^2}{8\pi\epsilon_0 r_{\text{hyd}}} \left(\frac{1}{\epsilon_{\text{op}}} - \frac{1}{\epsilon_r} \right)$$

MeCN ($\epsilon_r=37.5$, $\epsilon_{\text{op}}=1.81$); the production annotation uses a molecular-cavity nonequilibrium PCM, with Born as a check.

Report conventions explicitly:

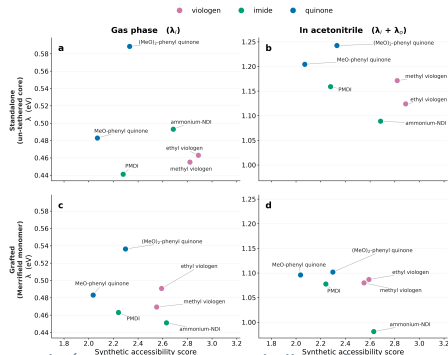
$$\lambda_{\text{het}} = (\lambda_O + \lambda_R)/2 + \lambda_{o,1};$$

$$\lambda_{\text{SE}}(d) = \lambda_O + \lambda_R + 2\lambda_{o,1}(1 - a/d).$$



At D3TaLES's own level, inner-sphere λ agrees to
median $|\Delta| = 0.09 \text{ eV}$ ($n = 322$).

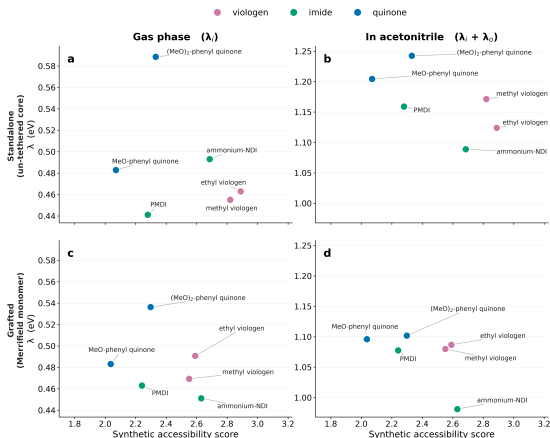
Candidates: SA vs. reorganization energy



All six are *grafted* model-compounds (redox core + 4-methylbenzyl monomer proxy); short class names label them.

- one point per molecule (mean over its charge states); the six cluster tightly ($\lambda_i \approx 0.45\text{--}0.54$ eV — similar across families).
- ammonium-NDI**: lowest $\lambda \Rightarrow$ fastest ET;
quinones highest.
- gas $\rightarrow$ MeCN: $+\lambda_o$ lifts each point $\sim 0.5\text{--}0.6$ eV (note per-panel y-zoom).
- λ_i validated (D3TaLES $R^2=1.0$); λ_o Marcus/Born estimate (r_{hyd} caveat) — read *relative* trends.

Functionalization $\times$ solvation: the 2×2



Rows = **before** (standalone core) vs **after** functionalization (grafted: redox core + 4-methylbenzyl monomer handle).

Cols = gas (λ_i) vs MeCN ($\lambda_i + \lambda_o$).

Functionalization barely moves λ_i — ± 15 – 50 meV, mixed sign:

- viologens: +15 / +29 meV
- PMDI +22; **NDI -44**; quinones 0 / -52

Solvation adds a near-uniform **+0.5–0.6 eV (λ_o)**.

$\Rightarrow$ the **redox core sets λ_i** ; the monomer handle is a small perturbation, so the cheap standalone λ is a good predictor of the grafted value.

What we learned (the durable lessons)

What works

- ranking (within charge class)
- xtb RRHO thermal (open-shell safe)
- disproportionation stability
- reorganization energy λ
- intact/bound state as a necessary-condition gate
- capacity (n , MW): exact

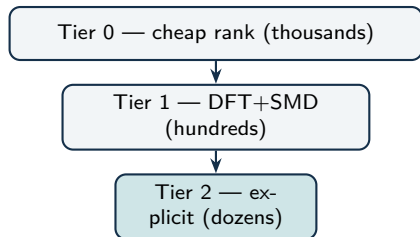
Negative results (saved effort)

- continuum ion-pairing: **refuted** (spacing off by ~ 11 V)
- explicit microsolvation (4 MeCN): **no gain** for viologen
- absolute redox offset: **unresolved** (OROP vs in-house anchors)
- dimerization absolute: **unreliable** (+2 desolvation)
- solubility / crossover: proxies only

Corroborated by the metabolic- ΔG pipeline: 0.14 eV via isodesmic cancellation $\Rightarrow$
cancellation, not more physics, is the lever.

Putting it together: a tiered, multi-objective search

Tiered funnel (explicit solvation is a finisher, never for thousands):



Pareto scorecard — built correctly

- objectives = **trustworthy** axes only (capacity, λ , disprop. stability, ranked potential)
- proxies (solubility, SA, crossover) = **filters**, not axes
- **integrity** = hard necessary-condition pre-filter
- missing applicable data = **INCOMPLETE**, never a favorable value
- split **anolyte** / **catholyte** (potential isn't "more is better")
- σ -aware domination (don't rank within error)
- read the front against the *raw* axes — no scalarized score (arbitrary weights, σ -blind, pool-relative)

Applying it: candidate scorecard → shortlist

σ -aware Pareto on the current rankable grafted set; rows missing an applicable objective are **INCOMPLETE** and cannot enter or dominate the primary front:

id	E_{an}	Cap	λ	ΔG_d	Par.
aq_benzyloxy [*]	−1.46	163	0.39	61	Y
pmdi	−1.46	160	0.47	64	—
mophquinone [*]	−1.02	176	0.52	92	Y
dmophquinone	−1.00	147	0.57	89	—
viologen [†]	−0.94	194	0.47	51	TBA
aq_benzylamino [‡]	−1.57	164	—	76	partial

Ordered by primary axis (E_{an}). E_{an} [V vs Fc], Cap [mAh/g], λ [eV], ΔG_d (disprop.) [kJ/mol].

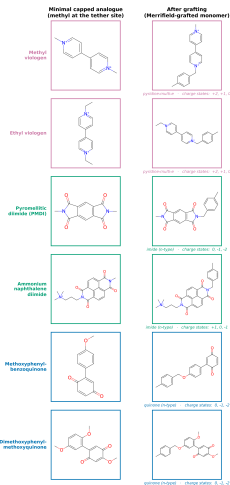
* = primary-front under both salts; † joins under TBAPF₆; ‡ is a partial front only (no QC-clean λ). The front and raw axes are reported without an arbitrary scalar score.

Completeness matters: a missing or QC-flagged λ cannot make a candidate look artificially fast. The row remains visible as a partial front, with its missing objective.

⇒ across this tight set voltage is *non-discriminating*; **capacity** + **disproportionation stability** + λ drive the ranking.

Current focus: six multi-electron starting candidates

Before vs. after grafting (six starting candidates)



Two **viologens**, two **imides**, and two **quinones**: capped core (left) versus Merrifield-grafted proxy (right).

Next steps

Done: protocol-addressed energies; unified σ -tagged scorecard; integrity and completeness contracts; σ -aware Pareto engine.

1. **Resolve the E° offset** — cross-run in-house and OROP molecules; audit OROP's thermodynamic cycle and reference convention before calibration.
2. **Broaden benchmarks** — second reductions and at least ten two-wave molecules.
3. **Complete kinetics** — conformer-matched core λ for tether jumps and solvated λ where a gas dianion is unbound.
4. **Polymer effects** — backbone/tether constraints & inter-monomer dimerization (where the *relative* dimerization ranking becomes key).
5. **Absolute accuracy** (if needed) — same-charge referencing to remove the per-class redox bias.

Summary

We can **rank** redox candidates and screen **stability**, **kinetics**, and **state integrity** on a physics-based footing — with each axis **honestly labelled** by how far we can trust it.

The σ -aware scorecard & Pareto engine are built and applied. Next: resolve absolute E° and close the incomplete kinetics rows before scaling.